# LARGER FEW-DISTANCE SETS IN $\mathbb{R}^6$, $\mathbb{R}^7$, $\mathbb{R}^8$ AND $\mathbb{R}^9$ FROM SHELLS OF ROOT LATTICES

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ABSTRACT. A finite set $X \subset \mathbb{R}^d$ is an $s$-distance set if exactly $s$ distinct distances occur between its points. Szöllősi and Östergård (Electron. J. Combin. 27 (2020), P1.23) tabulated the largest known $s$-distance sets in $\mathbb{R}^d$ for $d \le 8$ and $s \le 6$; for $s \in \{5,6\}$ and $d \in \{6,7,8\}$ the entries come from truncated simplices, and the authors remark that these are "rather weak". We show that the vectors of norm at most 6 of the root lattices $E_6$, $E_7$, $E_8$ contain much larger few-distance sets whose squared distances are $2, 4, \dots, 2s$. Subsets found by a maximum-clique local search have 133 and 227 points in $\mathbb{R}^6$ ($s = 5, 6$), 130, 253 and 388 points in $\mathbb{R}^7$ ($s = 4, 5, 6$) and 258, 520 and 910 points in $\mathbb{R}^8$ ($s = 4, 5, 6$), against the previously recorded 112, 140; 127, 168, 280; and 241, 252, 504. The same method gives 4- and 5-distance sets of 235 and 440 points in $\mathbb{R}^9$, where the best explicit constructions we are aware of have 210 and 360 points. For $s = 5$ the sets have the transparent form $\{0\} \cup R \cup S$, where $R$ is the root system and $S$ is a subset of the second shell with pairwise inner products at least $-1$; we prove a general lemma of this kind for every even lattice of minimal norm 2. All sets are given explicitly as integer vectors and are verified by an exact, dependency-free script.

## 1. INTRODUCTION

A set $X$ of $n$ distinct points in $\mathbb{R}^d$ is an *$s$-distance set* if the set $A(X) = \{\|x-y\| : x \neq y \in X\}$ of distances between its points has exactly $s$ elements; a 1-distance set is the vertex set of a regular simplex and has at most $d+1$ points. Determining the largest $s$-distance sets in $\mathbb{R}^d$ is a classical problem of discrete geometry, going back to Kelly [6] and Einhorn and Schoenberg [4] for $s = 2$. The general upper bound

$$|X| \le \binom{d+s}{s} \tag{1}$$

is due to Bannai, Bannai and Stanton [1] and Blokhuis [2], with refinements for spherical sets by Musin and Nozaki [8]; on the other side, for $d \ge 2s-1$ the $s$-element subsets of a $(d+1)$-element set, viewed as 0/1-vectors in the hyperplane $x_1 + \cdots + x_{d+1} = s$ of $\mathbb{R}^{d+1}$, form an $s$-distance set with $\binom{d+1}{s}$ points. Maximum 2-distance sets are known for $d \le 8$ [7], maximum planar $s$-distance sets for $s \le 6$ [5, 12, 11], and for $d \ge 3$, $s \ge 3$ the exact maxima are known only in a handful of cases: 12 for $(s,d) = (3,3)$ [10], and 13 for $(4,3)$ and 16 for $(3,4)$ [11].

Szöllősi and Östergård [11, Table 1] collected the best lower bounds known for $d \le 8$ and $s \le 6$; the part relevant here is reproduced in Table 1. Their entries for $s \in \{3,4\}$ and $d \in \{6,7,8\}$ come from the root system $E_8$ and its subconfigurations: the 240 roots and the origin form a 4-distance set of 241 points in $\mathbb{R}^8$, and 120 pairwise non-antipodal roots together with two further points form a 3-distance set with 122 points [11, Example 2]. The entries for $s \in \{5,6\}$ come from the vertices of truncated simplices, namely $(d-1)\binom{d+1}{2}$ points for $s = 5$ and $(d-2)\binom{d+1}{3}$ points for $s = 6$ [11, Lemmas 4 and 5]. The authors write that these last bounds "are rather weak" and that it is "plausible that they can be significantly improved".

In this note we confirm this expectation by a simple observation: instead of the root system alone one should use *several shells* of the root lattice. The vectors of the $E_8$ lattice of norm

TABLE 1. The lower bounds of [11, Table 1] for $s \geq 3$ and $d \geq 5$.

| $s\backslash d$ | 5 | 6 | 7 | 8 |
|---|---|---|---|---|
| 3 | 24 | 40 | 65 | 122 |
| 4 | 41 | 73 | 127 | 241 |
| 5 | 66 | 112 | 168 | 252 |
| 6 | 96 | 140 | 280 | 504 |

$2, 4, 6$ number 240, 2160 and 6720, and by integrality the squared distance between two lattice vectors of norm at most 6 is an even integer at most 24. Choosing a subset of these vectors in which only the squared distances $2, 4, \ldots, 2s$ occur is a maximum clique problem in an explicit graph, and a standard local search solves it well enough to beat every entry of Table 1 with $d \geq 6$ and $s \geq 5$, as well as the two 4-distance entries with $d \in \{7, 8\}$.

**Theorem 1.1.** *There exist $s$-distance sets in $\mathbb{R}^d$ of the following cardinalities, with squared distances $\{2, 4, \ldots, 2s\}$, all of whose points lie in the lattice $E_d$ ($d = 6, 7, 8$):*

| | $d = 6$ | $d = 7$ | $d = 8$ |
|---|---|---|---|
| $s = 4$ | (73) | 130 (127) | 258 (241) |
| $s = 5$ | 133 (112) | 253 (168) | 520 (252) |
| $s = 6$ | 227 (140) | 388 (280) | 910 (504) |

*The numbers in parentheses are the previously recorded lower bounds of Table 1. Moreover there exist a 4-distance set of 235 points and a 5-distance set of 440 points in $\mathbb{R}^9$, with squared distances $\{2, 4, 6, 8\}$ and $\{2, \ldots, 10\}$, all of whose points lie in the lattice $D_9$.*

For $(s, d) = (4, 6)$ we did not improve the value 73 attained by the 72 roots of $E_6$ and the origin. For $d = 9$ no table of lower bounds seems to exist in the literature; the two values are to be compared with the larger of the folklore bound $\binom{d+1}{s}$ and the truncated-simplex bound of [11], namely $\binom{10}{4} = 210$ for $s = 4$ and $\max\{\binom{10}{5}, 8\binom{10}{2}\} = 360$ for $s = 5$.

The sets are listed as integer vectors (twice the actual coordinates) in the ancillary file `certificate.json`; the script `verify_certificate.py`, reproduced in Appendix A, checks in exact integer and rational arithmetic that the points are distinct, that they lie in the stated lattice, that exactly the squared distances $2, 4, \ldots, 2s$ occur, and that the affine span has dimension exactly $d$. Two of the smaller sets are also printed in Appendix B.

The paper is organised as follows. Section 2 contains the elementary lemmas that explain why shells of an even lattice are a natural source of few-distance sets, in particular a clean description of the 5-distance sets of Theorem 1.1. Section 3 describes the computer search, the graphs involved, and the checks we performed against the known values. Section 4 records the structure of the sets found, including the spherical few-distance sets they contain. Section 5 collects remarks and open questions.

## 2. Few-distance sets in shells of an even lattice

Throughout, $L \subset \mathbb{R}^d$ is an *even integral lattice of minimal norm* 2: $\langle x, y\rangle \in \mathbb{Z}$ and $\|x\|^2 \in 2\mathbb{Z}$ for all $x, y \in L$, and $\|x\|^2 \geq 2$ for $x \in L \setminus \{0\}$. Examples are the root lattices $A_n$, $D_n$ ($n \geq 4$), $E_6$, $E_7$, $E_8$ and their direct sums; we refer to [3, Chapter 4] for the standard facts used below. We write $L_{2k} = \{x \in L : \|x\|^2 = 2k\}$ for the shells, so that $L_2 = R$ is the root system of $L$, and $L_{\leq 2K} = \bigcup_{k \leq K} L_{2k}$ for the ball.

**Lemma 2.1.** *Let $x \in L_{2a}$ and $y \in L_{2b}$ be distinct. Then*

$$\|x-y\|^2 = 2a + 2b - 2\langle x, y\rangle \in 2\mathbb{Z}_{>0}, \qquad |\langle x, y\rangle| \le 2\sqrt{ab}, \tag{2}$$

*with equality in the second relation only if $x = \pm y$. In particular $\|x-y\|^2 \le 2s$ if and only if $\langle x, y\rangle \ge a + b - s$.*

*Proof.* The first formula is the polarisation identity; $\|x-y\|^2$ is a positive even integer because $x - y \in L \setminus \{0\}$; the bound on the inner product is the Cauchy–Schwarz inequality, with equality only for parallel vectors, which in a lattice of minimal norm 2 means $x = \pm y$ when $a = b$ and cannot happen when $a \ne b$ unless $y$ is an integer multiple of $x$ (then $\sqrt{b/a} \in \mathbb{Z}$ and the inequality is strict). □

**Proposition 2.2.** *Let $X \subseteq L_{\le 2K}$ and let $s \ge 1$. Then $X$ is a $t$-distance set for some $t \le s$ with $A(X)^2 \subseteq \{2, 4, \dots, 2s\}$ if and only if*

$$\langle x, y\rangle \;\ge\; \tfrac{1}{2}\big(\|x\|^2 + \|y\|^2\big) - s \qquad \text{for all } x \ne y \in X. \tag{3}$$

*Consequently, the subsets of $L_{\le 2K}$ whose squared distances lie in $\{2, \dots, 2s\}$ are exactly the cliques of the graph $\Gamma(L, K, s)$ on $L_{\le 2K}$ in which two vectors are adjacent when their squared distance is at most $2s$; such a clique is an $s$-distance set if all $s$ values occur.*

*Proof.* Immediate from Lemma 2.1: all squared distances between points of $X$ are positive even integers, and they are at most $2s$ precisely under (3). □

Proposition 2.2 shows that for lattice sets the choice $A(X)^2 = \{2, \dots, 2s\}$ is the natural one: it is the only $s$-element set of positive even integers that is closed downwards, and downward closure is what turns the constraint on each pair into a one-sided inequality on the inner product. The next statement singles out the case that produces all our 5-distance sets.

**Lemma 2.3.** *Let $S \subseteq L_4$ be a set of norm-4 vectors with $\langle x, y\rangle \ge -1$ for all $x \ne y \in S$. Then $X = \{0\} \cup R \cup S$ is a $t$-distance set for some $t \le 5$ with $A(X)^2 \subseteq \{2, 4, 6, 8, 10\}$, and $|X| = 1 + |R| + |S|$. If all five values $2, \dots, 10$ occur (as they do in all examples below), then $t = 5$.*

*Proof.* We verify (3) with $s = 5$ for each type of pair. Two distinct roots satisfy $\langle r, r'\rangle \ge -2$, while (3) only asks for $\langle r, r'\rangle \ge 2 - 5 = -3$; explicitly $\|r - r'\|^2 = 4 - 2\langle r, r'\rangle \in \{2, 4, 6, 8\}$. For a root $r$ and $x \in S$, Lemma 2.1 gives $|\langle r, x\rangle| \le 2\sqrt{2} < 3$, hence $\langle r, x\rangle \ge -2 = 1 + 2 - 5$ and $\|r - x\|^2 = 6 - 2\langle r, x\rangle \in \{2, 4, 6, 8, 10\}$. For $x \ne y$ in $S$ the hypothesis $\langle x, y\rangle \ge -1 = 2 + 2 - 5$ gives $\|x - y\|^2 = 8 - 2\langle x, y\rangle \le 10$. Finally $\|0 - r\|^2 = 2$ and $\|0 - x\|^2 = 4$; and $0 \notin R \cup S$, $R \cap S = \emptyset$ because the norms differ, so $|X| = 1 + |R| + |S|$. □

Thus for the lattices $E_6, E_7, E_8$, whose root systems have 72, 126 and 240 elements, a 5-distance set with $1 + |R| + |S|$ points is obtained from any subset $S$ of the second shell in which no two vectors have inner product $-2$, $-3$ or $-4$, that is, in which no two vectors are at squared distance 12 or 16. Since $L_4$ lies on the sphere of radius 2, dividing by 2 turns the condition into a spherical-code condition: $S/2$ is a set of unit vectors with pairwise inner products at least $-1/4$, i.e. with pairwise angles at most $\arccos(-1/4) \approx 104.5°$.

**Corollary 2.4.** *Let $N_5(L)$ be the largest size of a subset of $L_4$ with pairwise inner products at least $-1$. Then $\mathbb{R}^d$ contains a 5-distance set of $1 + |R| + N_5(L)$ points. We found subsets of $L_4$ of sizes 60, 126, 279 and 295 for $L = E_6, E_7, E_8, D_9$ respectively (out of $|L_4| = 270$, 756, 2160 and 2034), giving the 5-distance sets with 133, 253, 520 and 440 points of Theorem 1.1.*

The second shell of $D_n$ consists of the $2n$ vectors $\pm 2e_i$ and the $16\binom{n}{4}$ vectors $\pm e_i \pm e_j \pm e_k \pm e_l$ $(i < j < k < l)$, which yields an explicit family for all $n$.

*Example* 2.5. Let $S_n \subset (D_n)_4$ consist of the $n$ vectors $2e_i$ and the $\binom{n}{4}$ vectors $e_i+e_j+e_k+e_l$ with $i<j<k<l$. All inner products within $S_n$ are non-negative, so by Lemma 2.3 $\{0\}\cup R(D_n)\cup S_n$ is a 5-distance set in $\mathbb{R}^n$ with $1+2n(n-1)+n+\binom{n}{4}$ points. For $n=9$ this gives 280 points, more than the folklore bound $\binom{10}{5}=252$ but less than the truncated-simplex bound 360; for every $n\geq 4$ the family is below the better of the two known bounds, since $\binom{n+1}{5}$ eventually dominates. The point of the example is rather to show what the search of Section 3 gains over the obvious choice of $S$: in $(D_9)_4$ it found a set of 295 vectors with pairwise inner products $\geq -1$, against the 135 vectors of $S_9$.

The analogous statements for $s=4$ and $s=6$ are less clean, because the whole root system can no longer be kept. For $s=4$, condition (3) requires $\langle r,x\rangle \geq -1$ for every root $r$ and every $x\in X\cap L_4$; but every vector $x$ of the second shell of $E_8$ is the sum $r_1+r_2$ of two orthogonal roots, and then $\langle -r_1,x\rangle=-2$, so admitting even one vector of the second shell forces the removal of roots. Our 258-point set in $\mathbb{R}^8$ keeps 123 roots and adds 134 vectors of norm 4. For $s=6$ vectors of the third shell can be used: for $z\in L_6$ and a root $r$ condition (3) asks for $\langle r,z\rangle\geq -2$, while $|\langle r,z\rangle|\leq\sqrt{12}<4$, so only roots with $\langle r,z\rangle=-3$ must be avoided; for $x\in L_4$ it asks for $\langle x,z\rangle\geq -1$, and within $L_6$ for $\langle z,z'\rangle\geq 0$. Table 3 shows how the search distributes the points among the shells.

## 3. The search

3.1. **Coordinates.** We use the standard coordinates of [3]. The lattice $E_8$ is the set of vectors of $\mathbb{R}^8$ whose coordinates are either all integers or all half-odd-integers and whose coordinate sum is even; $E_7$ is the sublattice of $E_8$ orthogonal to the root $e_1+e_2$, and $E_6$ the sublattice orthogonal to $e_1+e_2$ and $e_2+e_3$; $D_n$ is the set of integer vectors of $\mathbb{R}^n$ with even coordinate sum. To work with integers we double all coordinates, so that a vector of norm $2k$ becomes an integer vector of squared length $8k$ and squared distances are multiplied by 4. The shells have the following sizes [3, Chapter 4], which we recomputed:

| | $\lvert L_2\rvert$ | $\lvert L_4\rvert$ | $\lvert L_6\rvert$ |
|---|---|---|---|
| $E_6$ | 72 | 270 | 720 |
| $E_7$ | 126 | 756 | 2072 |
| $E_8$ | 240 | 2160 | 6720 |
| $D_9$ | 144 | 2034 | 7392 |

3.2. **The graphs.** For a lattice $L$, a bound $N\in\{4,6\}$ and $s\in\{3,4,5,6\}$ we form the graph $\Gamma=\Gamma(L,N/2,s)$ of Proposition 2.2 on the vertex set $L_{\leq N}$, in which $x\sim y$ if and only if $2\leq\|x-y\|^2\leq 2s$. Table 2 lists the number of vertices and the edge density $|E(\Gamma)|/\binom{|V|}{2}$. A clique of $\Gamma$ is a $t$-distance set with $t\leq s$; we discard cliques with $t<s$ or with affine dimension less than $d$ (neither happened for any reported set: for the sets of Theorem 1.1 every value $2,4,\dots,2s$ occurs many times, see Table 4).

3.3. **The clique search.** The graphs are dense and their cliques are large (hundreds of vertices), which is the regime in which exact maximum-clique algorithms are hopeless but stochastic local search is effective. We implemented a dynamic local search of the type introduced by Pullan and Hoos [9], in the following simplified form. The search maintains a current clique $C$ and, for every vertex $v$, the number $m(v)$ of members of $C$ not adjacent to $v$, and a penalty $p(v)$, initially zero.

(1) (*expand*) If some $v\notin C$ has $m(v)=0$, add to $C$ such a vertex with minimum penalty (ties broken at random).

TABLE 2. Vertices and edge densities of the graphs $\Gamma(L, N/2, s)$.

| $L$ | $N$ | $\lvert V\rvert$ | $s=4$ | $s=5$ | $s=6$ |
|---|---|---|---|---|---|
| $E_6$ | 4 | 343 | 0.698 | 0.883 | 0.961 |
| $E_6$ | 6 | 1063 | 0.396 | 0.583 | 0.703 |
| $E_7$ | 4 | 883 | 0.696 | 0.871 | 0.968 |
| $E_7$ | 6 | 2955 | 0.381 | 0.552 | 0.707 |
| $E_8$ | 4 | 2401 | 0.692 | 0.871 | 0.976 |
| $E_8$ | 6 | 9121 | 0.356 | 0.523 | 0.706 |

(2) (*plateau*) Otherwise, if some $v \notin C$ has $m(v) = 1$ and was not the vertex removed in the previous step, choose such a $v$ with minimum penalty, remove from $C$ its unique non-neighbour, and add $v$.
(3) (*perturb*) Otherwise increase the penalty of every member of $C$ by one, keep one random member of $C$ and remove all others; after every second perturbation, all positive penalties are decreased by one.

The best clique seen is recorded. With $10^6$ steps this takes a few seconds for $|V| = 2401$ and under a minute for $|V| = 9121$ on one core. Random seeds $1, 2, 3$ were used; for the graphs on $L_{\le 4}$ of $E_8$ all seeds returned the same clique sizes (258, 520, 817 for $s = 4, 5, 6$), and for the other graphs the sizes returned by different seeds differed by at most a few units. We therefore expect the sizes of Theorem 1.1 to be close to the clique numbers of the corresponding graphs, but we have not proved that they equal them. For $s = 6$ the third shell matters: on $E_8$ the size grows from 817 (norm $\le 4$) to 910 (norm $\le 6$), on $E_7$ from 364 to 388, and on $E_6$ from 163 to 227. For $s = 5$ nothing was gained from the third shell, in accordance with Lemma 2.3: every $z \in (E_8)_6$ has $\langle r, z\rangle = -3$ for some root $r$ (the third shell is a single orbit of the Weyl group, and $z = (2, 1, 1, 0, \dots, 0)$ has this property with $r = -e_1 - e_2$), so a vector of the third shell can only be added at the cost of roots. All code (`lat.py`, `run.py`, `dls.c`, `reproduce.py`) accompanies the paper.

3.4. **Checks against known values.** The same search applied to $s = 3$ returned exactly the recorded values 40, 65 and 122 of Table 1 in dimensions $6, 7, 8$ (the recorded 122-point set [11, Example 2] lies in $E_8$), and the recorded value 73 for $(s, d) = (4, 6)$. Applied to the lattice $D_5$ with vectors of norm at most 10, it returned exactly the recorded values 24, 41, 66 and 96 for $s = 3, 4, 5, 6$ in $\mathbb{R}^5$; in particular all entries of Table 1 for $d = 5$, which were obtained in [11] by a clique search over vectors with coordinates in a cube, are attained inside the lattice $D_5$. We regard these coincidences as evidence that the search finds cliques of the correct order of magnitude in these graphs, and that the lattice sets are the natural home of the known constructions.

For $d \le 5$ we found no improvements: on $D_4$, $D_5$, $A_4$, $A_5$, $\mathbb{Z}^4$ and $\mathbb{Z}^5$ with vectors of norm at most 10 the largest 5- and 6-distance sets found were at most the tabulated values, and for $d = 4$, $s \in \{5, 6\}$ the record sets (35 and 49 points, built from the 24-cell, its dual and their common centre, with distances involving $\sqrt{2 \pm \sqrt{2}}$) are not lattice sets and are strictly larger than the lattice cliques we found (31 and 40). For $d \in \{6, 7\}$ the lattices $A_6$, $A_7$ gave precisely the truncated-simplex values for $s = 5$ (112 and 176) and values near them for $s = 6$ (148 and 276 against 140 and 280), which is not surprising, since the truncated-simplex sets of [11, Lemmas 4 and 5] are subsets of shells of $A_d$; the lattices $E_6$ and $E_7$ are far better.

3.5. **Other distance sets.** We also ran the search with all $s$-element subsets of the possible squared distances $\{2, 4, \ldots, 16\}$ on $L_{\leq 4}$. In every case the downward-closed set $\{2, \ldots, 2s\}$ gave the largest cliques by a wide margin; for instance on $E_8$ with $s = 5$ the sets $\{2, 4, 6, 8, 12\}$ and $\{2, 4, 6, 10, 12\}$ gave cliques of 258 and 122 points (which are 4- and 3-distance sets, the largest distance being unused), and $\{2, 6, 8, 10, 12\}$ gave 54. This is in line with Proposition 2.2.

## 4. The sets

Table 3 gives the number of points of each norm in the ten sets of Theorem 1.1, and Table 4 the number of pairs at each squared distance. Several features are worth recording.

Table 3. Number of points of each norm in the sets of Theorem 1.1.

| $s$ | $d$ | $L$ | norm 0 | norm 2 | norm 4 | norm 6 | total |
|---|---|---|---|---|---|---|---|
| 4 | 7 | $E_7$ | 1 | 46 | 65 | 18 | 130 |
| 4 | 8 | $E_8$ | 1 | 123 | 134 | 0 | 258 |
| 4 | 9 | $D_9$ | 1 | 108 | 126 | 0 | 235 |
| 5 | 6 | $E_6$ | 1 | 72 | 60 | 0 | 133 |
| 5 | 7 | $E_7$ | 1 | 126 | 126 | 0 | 253 |
| 5 | 8 | $E_8$ | 1 | 240 | 279 | 0 | 520 |
| 5 | 9 | $D_9$ | 1 | 144 | 295 | 0 | 440 |
| 6 | 6 | $E_6$ | 1 | 56 | 90 | 80 | 227 |
| 6 | 7 | $E_7$ | 1 | 106 | 201 | 80 | 388 |
| 6 | 8 | $E_8$ | 1 | 202 | 512 | 195 | 910 |

Table 4. Number of pairs at each squared distance in the sets of Theorem 1.1 with $d \leq 8$.

| $s$ | $d$ | 2 | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|---|
| 4 | 7 | 1904 | 3373 | 2326 | 782 | | |
| 4 | 8 | 6094 | 14838 | 9506 | 2715 | | |
| 5 | 6 | 1587 | 2430 | 2760 | 1221 | 780 | |
| 5 | 7 | 4557 | 10338 | 9678 | 5535 | 1770 | |
| 5 | 8 | 15648 | 45915 | 42090 | 25605 | 5682 | |
| 6 | 6 | 3136 | 5855 | 7520 | 4540 | 3680 | 920 |
| 6 | 7 | 7566 | 19830 | 22246 | 15716 | 7844 | 1876 |
| 6 | 8 | 30384 | 108825 | 124482 | 100562 | 40146 | 9196 |

(1) All four 5-distance sets are of the form of Lemma 2.3: the origin, the full root system, and a subset $S$ of the second shell, with $|S| = 60, 126, 279, 295$. Within $S$ the inner products take exactly the values $-1, 0, 1, 2, 3$.
(2) In the 4-distance sets the origin is kept, roughly half of the roots are discarded, and the vectors of norm 4 that are kept have pairwise inner products in $\{0, 1, 2, 3\}$; in $\mathbb{R}^7$ the search also used 18 vectors of norm 6, with pairwise inner products in $\{3, 4, 5\}$.
(3) In the 6-distance sets the third shell contributes 80, 80 and 195 points in dimensions $6, 7, 8$; within the third shell the inner products are non-negative, as required by (3), and the vectors of norm 4 have pairwise inner products in $\{-2, \ldots, 3\}$.

(4) The origin belongs to every set, every set is contained in the ball of radius $\sqrt{6}$ about the origin, and the $s$ smallest possible squared distances $2, \dots, 2s$ all occur with large multiplicity.

4.1. **Spherical few-distance sets.** Removing the origin and restricting to one shell yields spherical few-distance sets, which are of independent interest (see [8] for the upper bounds and [11] for classification results in low dimensions). By item (1), the 279 vectors of norm 4 in the 520-point set form a spherical 5-distance set in $\mathbb{R}^8$ whose inner products, after normalisation to unit vectors, are $\{-1/4, 0, 1/4, 1/2, 3/4\}$; similarly the 126 vectors in $E_7$ and the 60 vectors in $E_6$ give spherical 5-distance sets in $\mathbb{R}^7$ and $\mathbb{R}^6$ with the same inner products. By item (2), the 134 vectors of norm 4 in the 258-point set form a spherical 4-distance set in $\mathbb{R}^8$ with inner products $\{0, 1/4, 1/2, 3/4\}$, and by item (3) the 512 vectors of norm 4 in the 910-point set form a spherical 6-distance set in $\mathbb{R}^8$ with inner products $\{-1/2, -1/4, 0, 1/4, 1/2, 3/4\}$. We have not attempted to optimise these spherical sets separately.

## 5. Remarks and questions

*Remark* 5.1 (Why root lattices). Proposition 2.2 applies to any even lattice of minimal norm 2, and Lemma 2.3 to any such lattice with a large root system and a dense second shell. Among lattices of dimension $d \in \{6, 7, 8\}$ the root lattice $E_d$ has the largest root system and the densest shells, which is why it wins for $s = 4, 5$ and, with the third shell, for $s = 6$. For $d = 9$ we used $D_9$; the lattice $E_8 \oplus A_1$ has the larger root system, with 242 roots, but a smaller second shell ($2160 + 2 \cdot 240 = 2640$ vectors, of which those involving the $A_1$ component have restricted inner products with the rest), and we did not explore it.

*Remark* 5.2 (Larger $s$ and $d$). Nothing in the method is specific to $s \le 6$ or $d \le 9$: for $s \ge 7$ the fourth shell enters, and in dimensions 10 to 24 the lattices $D_n$, $E_8 \oplus E_8$, the Barnes–Wall lattice and the Leech lattice (the last two suitably scaled) are candidates. For fixed $s$ and growing $d$, however, sets contained in a ball of fixed radius grow at most exponentially in $d$, whereas the folklore bound $\binom{d+1}{s}$ grows like $d^s/s!$; so the comparison in intermediate dimensions is a genuine question, which we leave open.

*Remark* 5.3 (Exactness). The reported cardinalities are clique sizes found by a randomised local search and are therefore lower bounds on the clique numbers of the graphs $\Gamma(L, N/2, s)$. For the graphs on at most 2401 vertices an exact computation of the clique number, exploiting the large automorphism group (the Weyl group of $E_8$ has order 696,729,600), may be feasible with the techniques of [11, 7]; it would settle, for instance, whether 520 is the largest 5-distance set of the form $\{0\} \cup R(E_8) \cup S$.

*Question* 5.4. The upper bound (1) gives 495, 1287 and 3003 for $(s, d) = (4, 8), (5, 8), (6, 8)$. What are the true maxima? In particular, is there an $s$-distance set in $\mathbb{R}^8$, $s \le 6$, larger than the largest clique in the graph on the vectors of $E_8$ of norm at most $2s$?

*Question* 5.5. What is the largest subset of the 2160 vectors of norm 4 of $E_8$ with pairwise inner products at least $-1$ (equivalently, the largest set of points of the second shell of $E_8$ with pairwise angles at most $\arccos(-1/4)$)? We found 279; by Lemma 2.3 each additional vector adds one point to the 5-distance set in $\mathbb{R}^8$.

## Appendix A. Verification script

The following Python 3 script uses only the standard library and takes well under a minute. It reads `certificate.json`, in which every point is an integer vector equal to twice the actual point.

```
#!/usr/bin/env python3
"""Standalone verification (Python 3 standard library only) of the few-distance sets in certificate.json.
Each point is given as an integer vector equal to twice the actual point; the sets live in the lattices E6, E7, E8
(coordinates in R^8, with E7 and E6 the sublattices orthogonal to (1,1,0,...,0) and to (1,1,0..),(0,1,1,0..) ),
or in the lattice D_9 (integer vectors of R^9 with even coordinate sum).
Checks: (1) points distinct, (2) exactly s distinct squared distances, equal to {2,4,...,2s},
(3) affine rank exactly d (exact rational Gaussian elimination), (4) lattice membership."""
import json, sys
from fractions import Fraction
def rank(rows):
    M=[[Fraction(x) for x in r] for r in rows]; rk=0; ncol=len(M[0]) if M else 0
    for c in range(ncol):
        piv=None
        for r in range(rk,len(M)):
            if M[r][c]!=0: piv=r; break
        if piv is None: continue
        M[rk],M[piv]=M[piv],M[rk]
        for r in range(len(M)):
            if r!=rk and M[r][c]!=0:
                f=M[r][c]/M[rk][c]; M[r]=[a-f*b for a,b in zip(M[r],M[rk])]
        rk+=1
    return rk
def in_e8(y):  # y = 2x; x in E8 iff all coords of x integers with even sum, or all half-odd-integers with even sum
    if all(t%2==0 for t in y): return (sum(y)//2)%2==0
    if all(t%2==1 for t in y): return sum(y)%4==0
    return False
cert=json.load(open(sys.argv[1] if len(sys.argv)>1 else 'certificate.json'))
ok=True
for key,c in sorted(cert.items()):
    s,d,n,S=c['s'],c['d'],c['n'],[tuple(p) for p in c['points_times_2']]
    assert len(S)==n and len(set(S))==n, "points not distinct"
    if c['lattice'].startswith('E'):
        assert all(len(p)==8 and in_e8(p) for p in S), "not in E8"
        if c['lattice']=='E7': assert all(p[0]+p[1]==0 for p in S)
        if c['lattice']=='E6': assert all(p[0]+p[1]==0 and p[1]+p[2]==0 for p in S)
    else:  # D_n: integer vectors with even coordinate sum
        m=int(c['lattice'][1:]); assert all(len(p)==m and all(t%2==0 for t in p) and (sum(p)//2)%2==0 for p in S), "not in D_n
    ds=set()
    for i in range(n):
        for j in range(i+1,n):
            ds.add(sum((a-b)**2 for a,b in zip(S[i],S[j])))
    ds4=sorted(x//4 for x in ds); assert all(x%4==0 for x in ds)
    assert ds4==[2*i for i in range(1,s+1)], (key,ds4)
    rk=rank([[a-b for a,b in zip(p,S[0])] for p in S[1:]])
    assert rk==d, (key,rk)
    print("%s: %d points in R^%d, squared distances %s, affine rank %d, previous record %d  -- OK"%(key,n,d,ds4,rk,c['previous
print("ALL CERTIFICATES VERIFIED" if ok else "FAIL")
```

## Appendix B. Two of the sets

Points are given as integer 8-vectors equal to twice the actual coordinates. The 133-point 5-distance set in $\mathbb{R}^6$ (all points orthogonal to $e_1 + e_2$ and $e_2 + e_3$):

```
(0,0,0,0,0,0,0,0) (-1,1,-1,-1,3,1,1,1) (0,0,0,-2,-2,-2,2,0) (1,-1,1,-1,1,-1,3,1) (0,0,0,2,-2,0,0,0) (1,-1,1,1,-1,-1,1,3) (0,0,
```

The 130-point 4-distance set in $\mathbb{R}^7$ (all points orthogonal to $e_1 + e_2$):

```
(0,0,2,0,0,0,0,-2) (-1,1,-1,-1,-1,-1,-1,1) (0,0,0,-4,-2,0,2,0) (0,0,0,0,0,0,0,0) (1,-1,-1,-3,-1,-1,1,1) (1,-1,1,-1,-1,-1,3,-1)
```

## References


[1] E. Bannai, E. Bannai, D. Stanton, *An upper bound for the cardinality of an s-distance subset in real Euclidean space, II*, Combinatorica **3** (1983), 147–152.

[2] A. Blokhuis, *Few-distance sets*, CWI Tract 7, CWI, Amsterdam, 1984.

[3] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, New York, 1999.

[4] S. J. Einhorn, I. J. Schoenberg, *On Euclidean sets having only two distances between points I–II*, Nederl. Akad. Wetensch. Proc. Ser. A **69** = Indag. Math. **28** (1966), 479–504.
[5] P. Erdős, P. Fishburn, *Maximum planar sets that determine k distances*, Discrete Math. **160** (1996), 115–125.
[6] L. M. Kelly, *Elementary problems and solutions. Isosceles n-points*, Amer. Math. Monthly **54** (1947), 227–229.
[7] P. Lisoněk, *New maximal two-distance sets*, J. Combin. Theory Ser. A **77** (1997), 318–338.
[8] O. R. Musin, H. Nozaki, *Bounds on three- and higher-distance sets*, European J. Combin. **32** (2011), 1182–1190.
[9] W. Pullan, H. H. Hoos, *Dynamic local search for the maximum clique problem*, J. Artificial Intelligence Res. **25** (2006), 159–185.
[10] M. Shinohara, *Uniqueness of maximum three-distance sets in the three-dimensional Euclidean space*, arXiv:1309.2047 (2013).
[11] F. Szöllősi, P. R. J. Östergård, *Constructions of maximum few-distance sets in Euclidean spaces*, Electron. J. Combin. **27** (2020), no. 1, P1.23.
[12] X. Wei, *A proof of Erdős–Fishburn's conjecture for* $g(6) = 13$, Electron. J. Combin. **19** (2012), no. 4, P38.

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